
Algorithm 1: End-to-End Pipeline of PSO-Optimized Random Forest for Multi-Class Mental Health Sentiment Classification

Input: Raw text dataset $D = \{(x_i, y_i)\}$, $y_i \in \{\text{Normal, Depression, Anxiety, Stress, Bipolar, Suicidal, Personality Disorder}\}$
Output: Optimal hyperparameters θ^* , best accuracy Acc^* , predicted labels $\hat{y}_{test}$

Phase 1: Data Preparation

Step 1: Text Preprocessing

foreach $x_i \in D$ **do**

- ┌ Convert text to lowercase;
- ┌ Remove punctuation, numbers, and special characters;
- ┌ Perform tokenization;
- ┌ Remove stopwords;
- ┌ Apply lemmatization;

Step 2: Feature Extraction

Transform preprocessed texts into feature vectors X using TF-IDF vectorization;

Step 3: Dataset Splitting

Split (X, y) into training set (X_{train}, y_{train}) and testing set (X_{test}, y_{test}) ;

Phase 2: PSO-Based Hyperparameter Optimization

Step 4: Swarm Initialization

Initialize swarm P with N particles;

Each particle represents a hyperparameter set θ_i ;

Initialize particle positions x_i and velocities v_i ;

Set $pbest_i \leftarrow x_i$, $gbest \leftarrow \arg \max(\text{Fitness})$;

Step 5: PSO Optimization Loop

while $iter < Max_Iterations$ **do**

foreach $particle\ i \in P$ **do**

- ┌ Decode x_i into θ_i ;
- ┌ Train Random Forest model M_i on (X_{train}, y_{train}) ;
- ┌ Predict validation labels $\hat{y}_{val}$;
- ┌ Compute fitness using Accuracy or F1-Macro;
- ┌ **if** $Fitness_i > Fitness(pbest_i)$ **then**
- ┌ $pbest_i \leftarrow x_i$;
- ┌ **if** $Fitness_i > Fitness(gbest)$ **then**
- ┌ $gbest \leftarrow x_i$;

foreach $particle\ i \in P$ **do**

- ┌ $v_i \leftarrow wv_i + c_1r_1(pbest_i - x_i) + c_2r_2(gbest - x_i)$;
- ┌ $x_i \leftarrow x_i + v_i$;

Phase 3: Final Modeling

$\theta^* \leftarrow gbest$;

Train final Random Forest model M^* on (X_{train}, y_{train}) ;

$\hat{y}_{test} \leftarrow M^*.predict(X_{test})$;

Compute final performance metrics (Accuracy, Precision, Recall, F1-score);

return θ^* , Acc^* , $\hat{y}_{test}$
